



# Bayesian source identification of urban-scale air pollution from point and field concentration measurements

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## Abstract

Air pollution poses a major threat to health, environment, and global climate. Characterizing the emission parameters responsible for air contamination can help formulate appropriate response plans. We propose an advanced methodology that uses Markov Chain Monte Carlo (MCMC) sampling within a Bayesian inference framework to invert for emission parameters of air contamination in an urban environment. We also use a high-resolution Lagrangian dispersion model to provide microscale wind computations as well as pollution concentration values in the presence of urban features with high complexity. Buildings and land use features were all integrated in a realistic urban setup that represents the region of King Abdullah University of Science and Technology, KSA. Boundary meteorological conditions acquired from a Weather Research and Forecasting (WRF) model simulation were employed to obtain the mesoscale wind field. We design numerical experiments to infer two common types of reference observations, a pollutant concentration distribution and point-wise discrete concentration values. The local  $L_2$  norm and global Wasserstein distance are investigated to quantify the discrepancies between the observations and the model predictions. The results of the conducted numerical experiments demonstrate the advantages of using the global optimal transport metric. They also emphasize the sensitivity of the inverted solution to the available observations. The proposed framework is proven to efficiently provide robust estimates of the emission parameters.

**Keywords** Bayesian inference · Air pollution dispersion modeling · Urban environment · Source term estimation · Optimal transport

## 1 Introduction

Anthropogenic emissions result from a variety of human actions, including conventional activities as well as accidental releases and leaks that may occur for various reasons, such as low maintenance, human error, terrorist attacks, and extreme natural events. Regardless of their origin, there is a dire need to identify and characterize the sources of such

contamination to enable the design and implementation of appropriate risk assessment and response strategies [1, 2]. This is particularly vital in urban environments where air quality and population health are closely interconnected [3–5]. However, several factors make the characterization of the source of an emission event a challenging task. The main challenge arises from the ill-posed nature of the underlying inverse problem and the non-uniqueness of the resulting solutions that can fit the available observations [6]. Another difficulty pertains to the uncertainties in the observations, which include noise from the sensing instruments and errors made during the data acquisition phase [7]. An additional key factor is the complexity of the physical dispersion mechanisms relating the source parameters to the observations. Such mechanisms highly depend on the meteorological, topographic, and urban features present at the meso- and micro-scales. Hence, uncertainties in the associated inputs affect the output of the model and eventually the inverted source parameters [8–11].

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Environmental monitoring provides different technologies to measure air quality and quantify air contamination. In remote sensing, recent advances have primarily been based on the optical characteristics of the atmospheric contaminants [12, 13]. Some of these techniques rely on passive sensing, which depends on the backscattering of solar radiation in the infrared and near-infrared regions. Other techniques use active sensing incorporating their own source of radiation, such as lasers emitted from light detection and ranging (LIDAR) or differential absorption LIDAR (DIAL) systems. The laser source can be deployed on the ground or mounted on satellites or aircrafts [14–18]. For both classes of remote techniques, air pollution data are obtained via the subsequent image-processing steps. These include image correlation coupled with ground measurements [19], the processing of images collected from camera–spectroradiometer calibration [20], or the processing of infrared images using high-pass and low-pass filters [21], as well as image contrast investigations and pattern recognition [22, 23].

Sensing networks are also being steadily developed to provide periodic measurements of air pollution with high spatiotemporal resolutions. Sensors are usually fixed at the ground level or can also be installed on top of an aircraft or a drone to measure contamination at varying horizontal and vertical locations in certain regions of interest [24, 25]. As a result of these environmental monitoring techniques, two different forms of observations are usually available. The first takes the form of a concentration field that represents a map with distributions of the average concentration values at the locations covered by that map. Such fields are obtained either from image-processing and remote-sensing techniques, or using measurements from well-established and rich sensing networks covering wide regions. Numerous interpolation techniques that estimate concentration fields from the monitoring stations measurements have been tested and reported in this context [26–29]. Conversely, in regions where such sensing networks are lacking, a finite number of deployed sensors may be available. Accordingly, the second form of observations comprises a set of discrete concentration values at some scattered locations.

Several methods have been proposed to identify the source parameters of air pollution events. The most intuitive one is backward tracking, in which a dispersion model is integrated backward in time from the observation to the emission [30]. In [31], this was implemented using the solution of the adjoint tracer transport and the Lagrange duality relation to characterize the source responsible for a large-scale radioactive tropospheric contamination event. Other approaches have applied optimization methods to seek for an optimal best-fit that minimizes a certain objective function measuring the fit between the model predictions and the observations. Some of them used the evolutionary genetic optimization algorithm by combining the forward dispersion model with

backward receptor models to derive certain source–receptor relationships [32]. Others have also employed regularization to enforce well-posed solutions [33–35]. The puff and Gaussian plume models have also been tested using synthetic data and have resulted in good solutions [32, 36].

Instead of searching for a single optimal combination of the source parameters, probabilistic approaches characterize a distribution of possible source parameter configurations [37–40]. This provides probabilistic information about the solution and hence allows for a quantification of its uncertainty, which cannot be achieved using the previously mentioned deterministic methods. This is often formulated as a Bayesian inversion problem, which aims to determine the probability distribution of the unknown parameters given the available measurements, so-called posterior [40]. Probabilistic approaches offer various advantages such as flexibility, using prior information, and obeying the likelihood principle [37, 38, 41, 42], in addition to accounting for the uncertainties in the observations and the model.

The Bayesian framework requires stochastic sampling tools that can fully assess the possible combinations of source parameters, hence quantifying their relative uncertainties [43, 44]. The most frequently used ones belong to the family of Markov Chain Monte Carlo (MCMC) algorithms [45]. This method has been already used to reconstruct emission sources from synthetic data [41] as well as actual measurements [37, 38]. It was also applied in [46] to an actual continental-scale accidental nuclear release based on measurements from an existing sensing network.

At the urban scale, inversion can be very challenging because of the complexity of the physical dispersion process in the presence of different urban features having heterogeneous geometries and densities [47]. For such environments, the Lagrangian dispersion model has been proven to perform well and lead to better results than those obtained by other models, such as Gaussian and Eulerian models [48–52]. Many studies have resorted to the Bayesian approach for source identification in urban environments using non-Lagrangian dispersion models. These include simple Gaussian puff models [45, 53, 54] or more complex models, such as the FEM3MP finite element code for flow and dispersion computations [55] and Gerris, a three-dimensional (3D) time-dependent flow code for incompressible Eulerian equations in complex geometries [56]. Despite the fact that such models can be run directly within the Bayesian framework to compute dispersion patterns [53–55], some studies have applied it indirectly to drive source–receptor relationships that can be used at different stages of the algorithm [37, 45, 56, 57]. A direct implementation of the Gaussian puff model was performed to identify area sources in an industrial park in China [53]. In [54], a similar model was used to characterize point sources in urban settings after calculating the puff backward trajectory leading to the sensor location with the

goal of narrowing the prior of the source location and achieving fast convergence. Fast performance can also be achieved by making use of a precomputed reusable database of the dispersion simulation results [55]. Even though urban environments usually extend over large areas and host complex features, investigated areas have ranged from small domains with less than 10 obstacles [54, 56] to relatively large domains with close to 100 obstacles [53, 55].

Previous studies have mostly relied on point-wise synthetic data and sensor readings, i.e., observations of the previously defined second form (discrete scattered concentration values) [37, 53, 55]. The observations have been obtained from databases that were constructed based on field experiments that mimic urban settings, such as the Mock Urban Setting Test (MUST) [58] and the Fusion Field Trails 2007 (FFT07) experiment [45]. In terms of algorithms implementation, the considered Bayesian frameworks merely used local dissimilarity metrics to measure the mismatch between the observations and the model outputs.

Herein, we propose a more advanced solution to the ill-posed air pollution source identification problem following a Bayesian inference framework along with MCMC sampling. We use a fully realistic high-resolution Lagrangian atmospheric dispersion model to provide microscale wind computations as well as pollution concentration values in the presence of urban features with high complexity. We apply this framework in a complex urban environment with an area of 17 km×16 km containing a total of 2,111 buildings. We then test synthetic observations of the two different types: concentration fields and point-wise discrete concentration readings. The proposed algorithm quantifies the dissimilarity between the modeled outputs and the reference observations according to a standard local metric and to the global Wasserstein distance [59, 60]. The main novelty of this paper is threefold:

- i. the implementation of a Bayesian inference framework for pollution source identification with a fully realistic urban modeling framework,
- ii. the introduction of the global Wasserstein distance as a dissimilarity measure in the likelihood,
- iii. and the inference of pollutant concentration data in the form of a distribution or a field, and not only as discrete values.

The study is structured as follows. Section 2 introduces the dispersion model along with the studied domain and the reference synthetic observations. Section 3 provides a detailed description of the algorithm and its Bayesian formulation. Section 4 outlines the numerical experiments that were conducted to assess the inverted solution with respect to the reference observations and used metrics. Finally, Section 5

discusses the significance and interpretations of the experimental results.

## 2 Model description

We employ the Graz Mesoscale Model-Graz Lagrangian Model (GRAMM-GRAL [61, 62]), which is an air pollution model composed of GRAMM, a mesoscale wind field model, and GRAL, a microscale wind field and Lagrangian particle transport model. It has been configured to model the concentrations of various contaminants at the urban and meso-scales emitted from diverse sources throughout several entire European cities [51, 63–66]. It was also used as a tool to simulate pollution in different urban geometries, such as highways with complex terrains, and specific streets intersections; as well as to study climate change mitigation in urban mobility [67–69]. Different studies validated the model by using several combinations of meteorological data, emission patterns and urban features to prove its compliance with the standard quality objectives and performance criteria [62, 70]. The validation covered the general model properties, such as convergence, grid characteristics and computational features. It also relied on many test cases of modeled data against real time measurements of wind conditions and pollutants concentrations. The model proved to comply with the German Guideline VDI 3783 (2015) and the guidelines of the Forum for Air quality Modelling in Europe (FAIRMODE) [51, 62, 70]. GRAMM-GRAL was also put through model-to-data and model-to-model comparisons. The GRAMM and GRAL models were compared to two widely used models in Australia: CALMET and CAL3QHCR. It was reported that GRAMM-GRAL produced outcomes whose quality is as good as those produced by the other models [71].

In our model implementation, two uniform rectilinear nested grids are considered: a low-resolution grid covering a larger domain for GRAMM and a high-resolution grid focusing on the domain of interest for GRAL. The model computes the flow and turbulence fields around the buildings for a specific topographic and urban setup by solving the Navier–Stokes equation along with the  $k$ - $\epsilon$  turbulence closure model. Then, based on the resulting predictions, GRAL computes the air pollution concentration field based on the defined sources. More details about the used parametrization schemes can be found in the model documentation in [61, 62]. As inputs, the model requires data that describe the domain, including the topography and surface features (such as the albedo, emissivity, and heat conductivity), shape files of the buildings and their characteristics, and the prevailing wind conditions (such as the speed, direction, and atmospheric stability class). Moreover, the particle transport model requires as input the locations, emission rates, and stack properties of any localized pollution sources.

The domain of the inference problem is set to be the campus of King Abdullah University of Science and Technology (KAUST) in Thuwal, Saudi Arabia, which is located on the coast of the Red Sea, 80 km north of Jeddah (Fig. 1). The domain was georeferenced using the software graphical user interface in the Universal Transverse Mercator (UTM) coordinate system of latitude and longitude. In this coordinate system, the study domain lies in the UTM zone 37N. Both the GRAMM and GRAL grids are uniformly meshed Cartesian grids such that the outer GRAMM domain has a resolution of 100 m and the inner GRAL domain has a fine resolution as high as 4 m. The topography and land use were defined based on input files extracted from digital elevation models of the ground elevation and coordination of information on the environment values [72]. Another major component are the manmade structures, which were imported via a shape file containing the 2,111 KAUST buildings, including their two-dimensional (2D) configurations, dimensions, and elevations. This shape file was manually and accurately created from the original blueprints of KAUST buildings.

The wind boundary conditions are obtained from high resolution data generated using a regional configuration of the Advanced Weather Research and Forecasting (WRF) model [73]. WRF was configured with three two-way interactive nested domains, with horizontal resolutions of 9 km, 3 km, and 0.6 km, and 60 vertical levels throughout the

atmospheric column. Suitable parametrization schemes were implemented for the microphysical, surface, and boundary layer processes as well as long-wave and short-wave radiation [74–76]. The model uses the United States Geological Survey topography, land use, and soil types, in addition to the Global Forecasting System analysis available at  $0.25^\circ$  grid produced by the National Centers for Environmental Prediction (NCEP) for initialization and boundary conditions (updated every 3 hours). WRF simulations were performed over 3 years between 2016 and 2018, using a consecutive integration method with daily initialization [77–79]. 36-hours simulations were conducted, starting at 12:00 UTC each day; the first 12 hours period was neglected as a spin up and the remaining 24 hours period was used to generate a 3 year sequence of very high-resolution (600 m) data for the region. The quality of this data was evaluated in [80] through extensive comparisons against different observations. GRAMM boundary conditions, including wind speed, wind direction, and atmospheric stability classes, were extracted from the WRF data at time intervals of 30 minutes. These are linearly interpolated by GRAMM to match its time stepping.

In our implementation, GRAMM is decoupled from GRAL such that the wind fields are precomputed by GRAMM before running the algorithm, stored, and used as a fixed database. Then, at every iteration, GRAL calls these



**Fig. 1** The geographic location of the study area KAUST

wind fields to predict the air pollution concentrations resulting from a contamination source (see Fig. 2). The geographic locations of the model domains are described in Section 3.4 and represented in Fig. 3. In our case, a single point emission source was defined according to its 2D location in the UTM coordinate system, stack height, type of released contaminants, emission rate, and emission duration. The simulated concentration fields for the reference source parameters were used as the synthetic observations, and were obtained based on the flow conditions represented in Fig. 4. We extracted two forms of synthetic reference observations: a concentration field of data available with a resolution of  $40\text{m} \times 40\text{m}$ , and concentration point-wise values (Fig. 5). To mimic real-

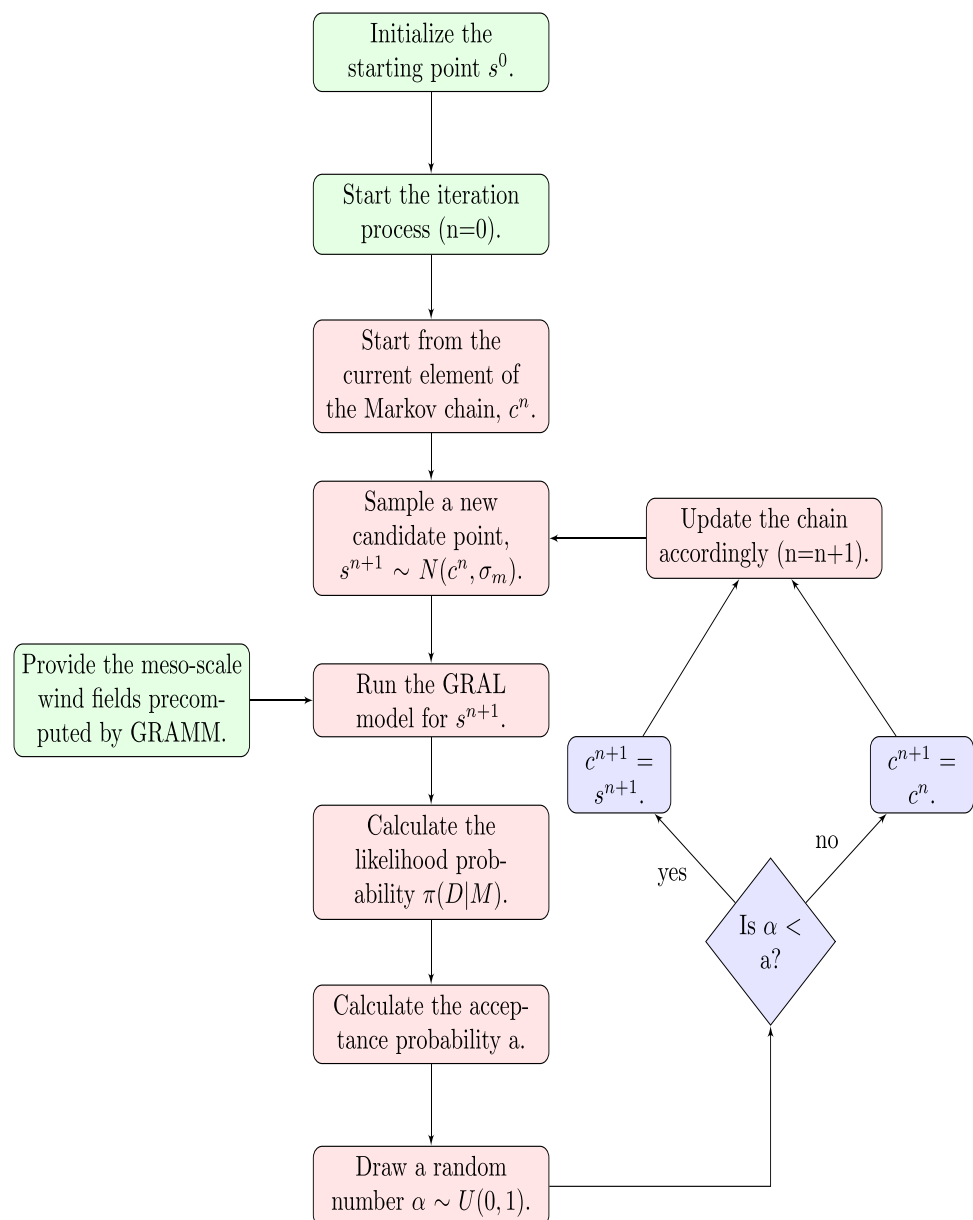
istic conditions, random noise was added to each of these observations (Section 4).

### 3 Inference methods

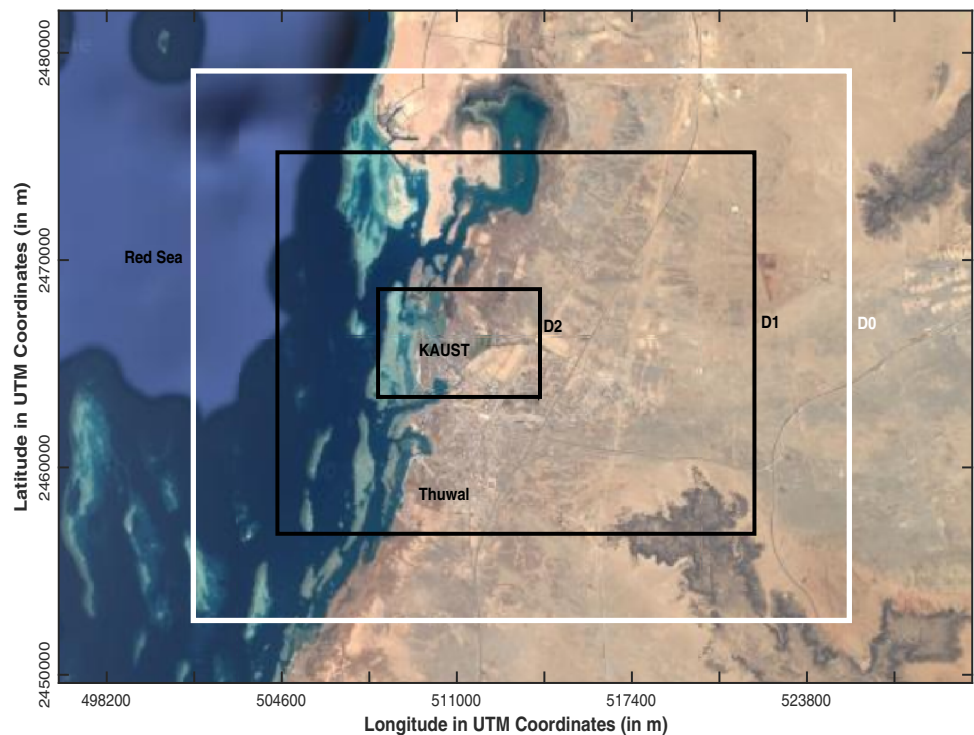
#### 3.1 Bayes' inversion

Bayes' rule states that the posterior probability is proportional to the product of the prior and the likelihood probabilities [81]. The prior represents knowledge about the model parameters prior to observing the data; while the likelihood expresses how likely the model predicts the observation

**Fig. 2** An overview workflow chart of the proposed algorithm



**Fig. 3** Geographic locations of the computational model domains. The white domain  $D_0$  is the GRAMM domain and is only used before running the algorithm. The black domains  $D_1$  and  $D_2$  are the two domains used by GRAL within the algorithm



given a certain set of model parameters. The Bayes' rule is generally expressed as

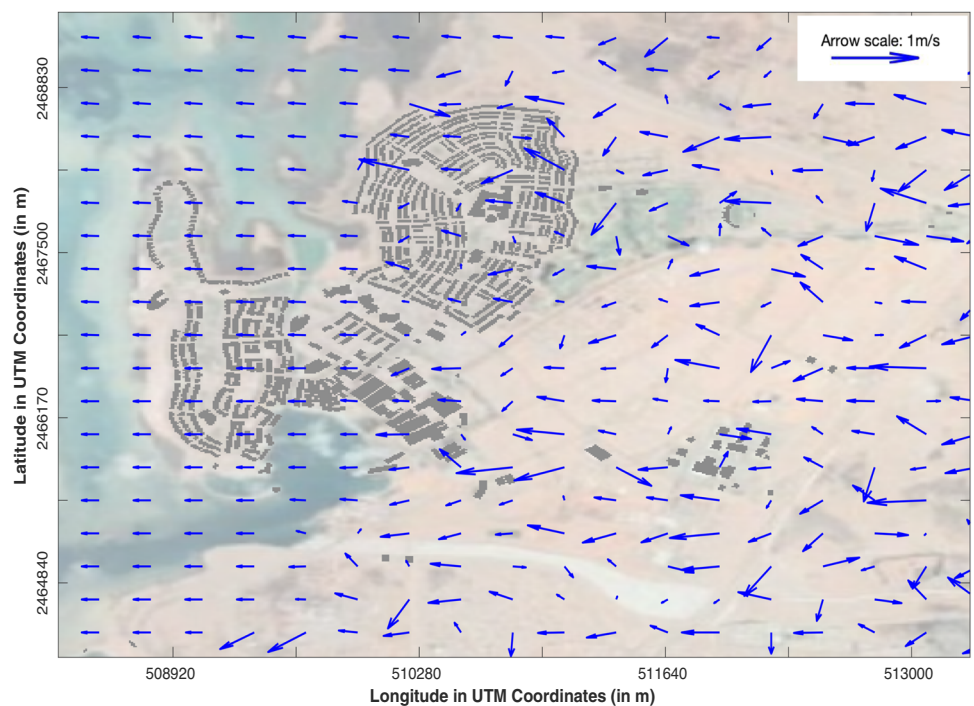
$$\pi(M|D) \propto \pi(M)\pi(D|M), \quad (1)$$

where  $M$  refers to the model parameters,  $D$  refers to the data,

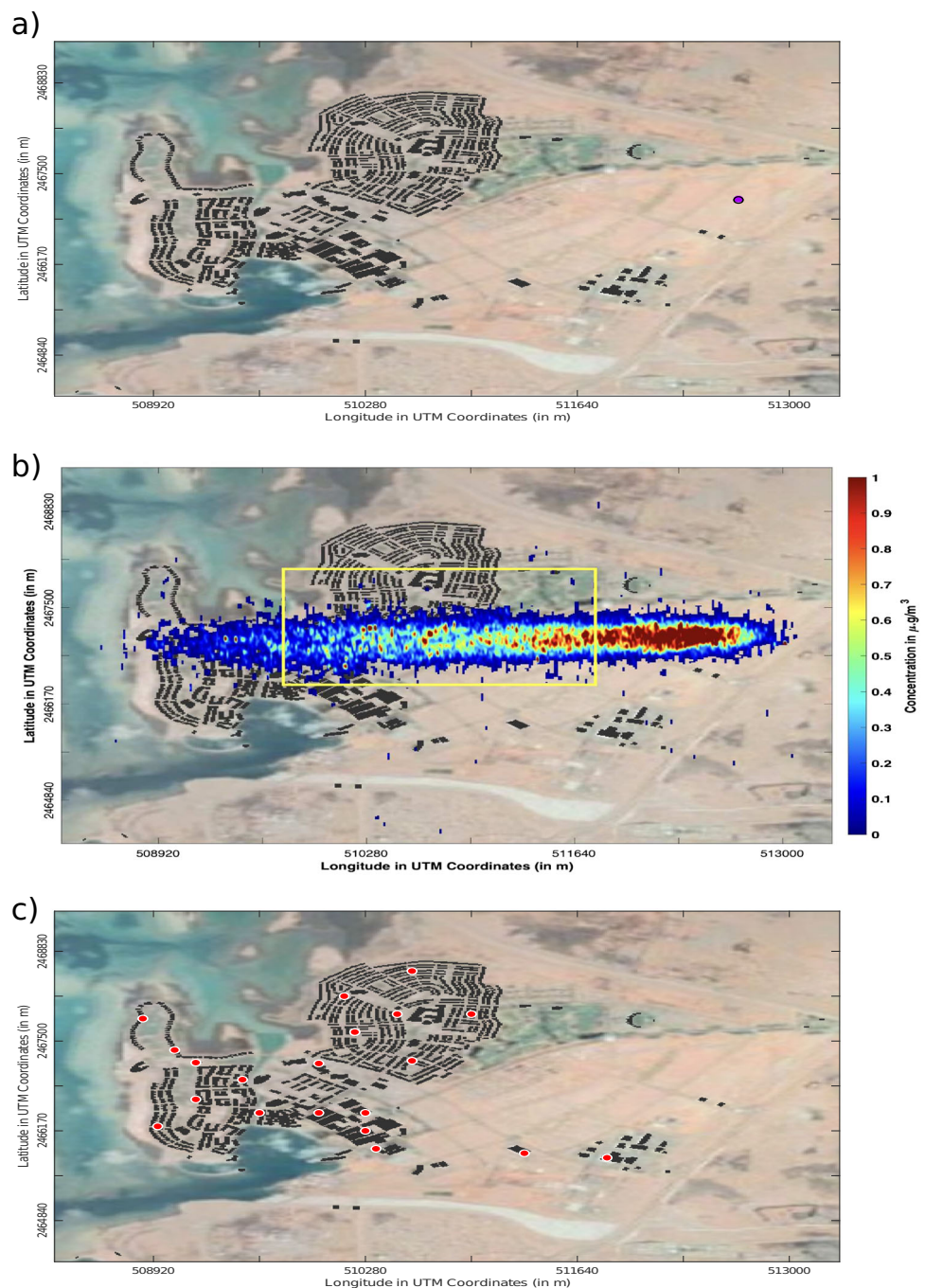
$\pi(M)$  is the prior,  $\pi(D|M)$  is the likelihood, and  $\pi(M|D)$  represents the posterior.

The model parameters in our case are the source location (the latitude and longitude equivalent to the  $x$ - and  $y$ -coordinates), the stack height  $h$ , the emission rate  $q$ , and the emission duration  $d$ . Here, we consider that no information

**Fig. 4** The wind conditions used in the experimental scenario to create the synthetic observations. The black polygons show the King Abdullah University of Science and Technology (KAUST) buildings



**Fig. 5** Complete synthetic concentration field obtained from the model output at an elevation of  $h = 10$  m, and at a time  $T = 6$  h. The black polygons show the King Abdullah University of Science and Technology (KAUST) buildings. a) The purple dot marks the true source location. b) The normalized synthetic concentration field used as a reference observation in the experiments is delineated by the yellow rectangle. c) The red scattered dots represent the pseudorandom locations of the sensors



is available about the emission parameters, so that an uninformative prior, i.e., a uniform distribution over predefined bounds for each of the five parameters, is used.

$$\pi(M_i) = \begin{cases} \frac{1}{a_{M_i} - b_{M_i}} & \text{if } M_i \in [a_{M_i}; b_{M_i}]; i = 1, \dots, 5 \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

Equation 1 then reduces to

$$\pi(M|D) \propto \pi(D|M), \quad (3)$$

indicating that the posterior is directly proportional to the likelihood within the prior bounds represented in Table 3. Based on the posterior distribution, different estimators of  $M$  can then be calculated along with their uncertainties. This includes classical estimators such as the posterior mean, which is based on a quadratic loss function, and the maximum a posteriori (MAP), which is based on the zero-one loss function [82]. Here, we take the *MAP* as the best solution.

### 3.2 The likelihood

The likelihood probability  $\pi(D|M)$  describes how well the model prediction matches the observation. In this context, for point-wise observations with  $n_p$  concentration values, various expressions of likelihood probability distributions have been used in different atmospheric modeling studies [37, 45]. Based on the principle of maximum entropy, the most conservative choice is a Gaussian likelihood expression [83]. For field observations, we assume that the likelihood distribution decays exponentially with the distance. We further consider that the observations are independent so that their likelihood probabilities of point-wise values and concentration fields, respectively, are products of  $n_f$  exponential terms:

$$\pi(D|M)_{points} = \prod_{k=1}^{n_p} \frac{1}{\sqrt{2\pi}\sigma_o} \exp^{\frac{-1}{2\sigma_o^2} distance^2}, \quad (4.1)$$

$$\pi(D|M)_{fields} = \frac{1}{\beta} \exp^{\frac{-distance}{\beta}}, \quad (4.2)$$

where  $n_f$  is the number of field observations,  $n_p$  is the number of point observations,  $\sigma_o$  is the standard deviation of the observational error, and  $\beta$  is a scaling parameter.

Suitable distances are chosen depending on the type of the observations.

a) *Local Point-Wise Difference* ( $\chi$ ): This distance is equivalent to the  $L_2$  norm and is used in the case of discrete observations and represents the squared value of the difference between  $n_p$  observations  $P_i$  and the corresponding model predictions  $C_i$ .

$$\chi = \sqrt{\sum_{i=1}^{n_p} (P_i - C_i)^2}. \quad (5)$$

b) *The Wasserstein Distance* ( $\delta$ ): This distance is inspired from the optimal transport theory where it is used to measure misfits between two functions or fields while enforcing certain conservation properties as constraints [84, 85]. The theory was developed by Gaspard Monge to minimize the cost needed to move a certain mass of sand from an initial location (a quarry) to a new location (a construction site) [86]. Mathematically, this cost is the product of the mass to be moved and the distance to be traveled while respecting the conservation of mass constraint [87]. Monge's formulation may encounter difficulties at the level of the conservation constraint and the uniqueness of the solution [88]. To deal with this, the modern optimal transport theory uses a convex relaxation of the initial Monge problem and introduces probability measures to minimize the cost function over a set of transport plans between different distributions [88]. This enabled the application of the optimal transport theory to

multiple domains, including machine learning, fluid dynamics, statistics, optics, probabilistic studies, shape analysis, and computer science, e.g. [87, 89–93].

For our problem, we treat each concentration field as a density that can be displaced. We use optimal transport to quantify the distance between the simulated and observed distributions, by minimizing the path required to translate one to the other. The minimization process expresses the dissimilarity as

$$\delta_{(f_0, f_1)} = \sqrt{\min_{T^* \in T(f_0, f_1)} J(T)}, \quad (6)$$

$$\text{with } J(T) = \int L(u, v) f_0(u) du.$$

where  $\delta_{(f_0, f_1)}$  denotes the Wasserstein distance between two distributions  $f_0$  and  $f_1$ ,  $T$  is the set of sufficiently regular bijections mapping  $f_0$  to  $f_1$ ,  $T^*$  is the unique bijection that minimizes  $J$ , and  $L$  is a symmetric cost function to move a unit from point  $u$  to point  $v = T(u)$ . The Euclidean distance is commonly used for  $L$  [94, 95].

Wasserstein distance was already used in [60] to quantify the mismatch between the fields of accidentally released radioactive elements in the Fukushima Daiichi accident in Japan, in order to avoid the double-penalty effect associated with local metrics. The latter occurs because a local distance can be highly sensitive to slight temporal or spatial differences between fields, making the corresponding inverse problem particularly challenging. We computed the Wasserstein distance using the code provided in [60], which adopts the fluid mechanic formulation of the optimal transport problem [59].

### 3.3 Sampling the posterior

To sample the posterior, we use the well-known Metropolis Hastings (MH) algorithm, which operates iteratively as follows [96].

1. Initialize the starting point  $s^0$  by randomly sampling a value for each of the parameters to be inferred according to the prior distributions.
2. *Iteration process.*
  - (a) Denote the current element of the Markov chain by  $c^n$ ; sample a new candidate point  $s^{n+1}$  from a normal distribution centered at  $c^n$  and having a standard deviation  $\sigma_m$ .

$$s^{n+1} \sim N(c^n, \sigma_m). \quad (7)$$

- (b) Run the GRAL model for the sampled set of source parameters  $s^{n+1}$  to predict the air pollution concentration.
- (c) Calculate the likelihood probability  $\pi(D|M)$  by evaluating the distance between the reference observation and the prediction obtained from the current sample of source parameters,  $s^{n+1}$ .
- (d) Calculate the acceptance probability, given by

$$a = \min \left( 1, \frac{\pi(M_{n+1}|D)}{\pi(M_n|D)} \right). \quad (8)$$

Then, draw a random number  $\alpha$  from the uniform distribution over the interval  $[0,1]$ .

$$\alpha \sim U(0, 1). \quad (9)$$

Compare  $a$  and  $\alpha$  to update the Markov chain accordingly.

$$c^{n+1} = \begin{cases} s^{n+1} & \text{if } \alpha < a, \\ c^n & \text{otherwise.} \end{cases} \quad (10)$$

The described MH algorithm was coded in Matlab. Our codes incorporate all the necessary functions for sampling, calling GRAMM-GRAL model, quantifying the dissimilarity and evaluating the posterior. An overview workflow chart of the proposed algorithm is represented in Fig. 2.

### 3.4 Computational aspect

Based on the MH algorithm, the proposed Bayesian framework requires numerous model predictions involving the GRAL model, which could be computationally very demanding. Note that GRAMM model imposes no computational costs on the algorithm because it is used once to simulate the wind fields before starting the iterative process. In this context, running the GRAL model over a smaller domain could significantly reduce the required computations. Accordingly, we split the model runs into two domains, which are selected such that the reference observations are included in each of them:  $D1$ , whose geographic boundaries are the same as the prior bounds and are wide enough (17 km×16 km) to compound the entire plume (Fig. 3); and a smaller  $D2$ , of dimensions 6 km×5 km. The two domains use the same grid resolution; however,  $D2$  includes significantly fewer

grid cells than  $D1$ . Because we aim to invert for the emission source parameters, the algorithm determines the location of the proposed point source with respect to the defined domains. If a sampled location falls inside  $D1$ , the model is integrated over  $D1$ ; otherwise, it is integrated over  $D2$  only. The computational costs required by the algorithm are shown in Table 1, and an example of the total run values and the computational saving is provided in Table 2. In each of the two domains, the time per iteration is calculated as a sum of the computational time required to complete a single model run, and to estimate the distance (refer to Table 1). Also, the percentage of total runs in each domain is averaged over all the obtained chains in an experiment. The size of the computational domain is based on the prior bounds outlined in Table 3. Significant computational savings are achieved because the run-time required for running the model in  $D2$  is one third of that required for  $D1$ . Note that more small subdomains may be also set within the prior bounds, but wide enough to cover the simulated plume. Here, we demonstrate the advantages of dividing the model runs into two subdomains.

## 4 Experimental design

The reference experimental scenario considers a point pollution source located to the east of KAUST campus, at  $x = 513,295$  m and  $y = 2,467,205$  m. This source is placed at a stack height of  $z = 10$  m above sea level and emits  $PM_{2.5}$  for a duration of  $d = 240$  min at a constant emission rate of  $q = 100$  kg/h. We note that our framework is very flexible and can be technically used for intermittent release cases, or any other scenarios. The extracted observations are perturbed by an unbiased normally distributed observational error with a standard deviation of  $500 \mu\text{g}/\text{m}^3$ , which is of the order of the mean pollutant concentrations over the domain (see Fig. 5). The scale parameter of the exponential likelihood distribution,  $\beta$ , is a constant equal to 40. This value maximizes the likelihood between the reference observation and the model prediction of the true source parameters with different realizations of the observational error.

The prior bounds and the settings of the performed source identification experiments are outlined in Tables 3 and 4, respectively. Exps. 1 and 2 were designed to assess the more common form of observation, the point-wise concentration

**Table 1** Analysis of the computational time required by the algorithm

| Computed quantity           | Computational time required per sample [s] |
|-----------------------------|--------------------------------------------|
| Concentration Field in $D1$ | 60                                         |
| Concentration Field in $D2$ | 20                                         |
| Wasserstein Distance        | 20                                         |

**Table 2** Example of the achieved computational time reduction in Experiment 3. The first two rows state the computational time for each of the two domains. The last three rows represent the total run time and computational savings of the algorithm

|                                                  | Inside D1         | Inside D2 |
|--------------------------------------------------|-------------------|-----------|
| Time required per iteration [s]                  | 80                | 40        |
| Average percentage of total runs (%)             | 52                | 48        |
|                                                  | Overall Algorithm |           |
| Original time required per 1000 iterations [h]   | 22.22             |           |
| Time required after splitting the model runs [h] | 16.88             |           |
| Percentage reduction in time (%)                 | 24                |           |

values. Here, the observation comprises 20 concentration values that represent the readings of 20 sensors mounted on different KAUST buildings (Fig. 5). Exp. 1 uses the local metric  $\chi$ , in contrast to Exp. 2 that treats the point-wise values as an incomplete field and uses the global metric  $\delta$ . This is done by generating a fictitious grid over the domain and assuming that each sensor is located at the nearest grid center. Then, the point-wise values are set at their corresponding sensors locations and zeros elsewhere. This is implemented for both, the observations as well as the model predictions, and use the Wasserstein metric such that zero concentrations do not contribute to the overall distance (see Eq. 6).

Exps. 3, 4, and 5 invert for concentration fields. Exp. 3 inverts for one field observation available at an elevation  $h = 10$  m and time  $T = 6$  hours, whereas each of Exp. 4 and 5 simultaneously invert for two field observations at two different elevations ( $h_1 = 10$  m;  $h_2 = 25$  m) and a given time ( $T = 6$  hours) in Exp. 4, and at a fixed elevation ( $h = 10$  m) and two different instants in time ( $T_1 = 6$  hours and  $T_2 = 8$  hours) in Exp. 5.  $T_0 = 0$  corresponds to the start of the emission process. We choose our reference observations to be collected at some time after the emission process has stopped (after  $T = 4$  hours); otherwise, identifying the source would be trivial just by visually inspecting the plume. Note that the selected time is also long enough for the released pollutants to reach the urban area under study. In addition, we assume that the field observations for each experiment cover only a part of the urban area of the investigated domain, as represented in Fig. 5. The goal of these experiments is to investigate the effect of additional data on the accuracy of the inferred solution.

Finally, Exp. 6 infers both point-wise and field observations to evaluate the effects of combined information. The experiment inverts for point-wise values available at an elevation of  $h_p = 10$  m and one field observation at an elevation of  $h_f = 25$  m; both available at time  $T = 6$  hours.

## 5 Results and discussion

The proposed Bayesian inversion framework was implemented based on the experiments outlined in Table 4. Several

MCMC simulations were performed in which a total of 100,000 samples were generated for each experiment. Each simulation is initialized by a unique starting point and produces 20,000 samples as shown in Fig. 6. Using two computational domains,  $D1$  and  $D2$ , as described in Section 3.4, resulted in a 25% saving in the computational cost compared to using only  $D1$ . Hereafter, the inference results using the  $L_2$  norm are first compared against the Wasserstein distance by analyzing the double penalty effect. Then, the efficiency of treating the point-wise observation as an incomplete field and using the global metric is evaluated. Finally, the sensitivity of the inverted solution to the observations used in the different experiments (Table 4) is analyzed. It is worth noting the probabilistic framework of interpreting the resulting posterior distributions and their characteristics in what follows.

### 5.1 The double-penalty effect

The Markov chains were sampled with an acceptance rate of 37% and 32% in Exps. 1 and 3, respectively. The chains indicate good mixing over the entire parameter space, as observed by the fluctuations in the trace plot of the chains for Exp. 3 (Fig. 6). After discarding the first 10% of the samples as burn-in, the posterior probability distributions were extracted using the kernel density estimation [97–99] and the results are presented in the corner plots of Figs. 7 and 8. The off-diagonal subplots represent the joint posterior probability for each pair of parameters, while the diagonal subplots show the marginal posterior probability distribution of each single parameter. The red lines indicate the true values of each inferred parameter in the subplots. Based on the resulting

**Table 3** Bounds of the uniform prior probability distributions of the five parameters of interest

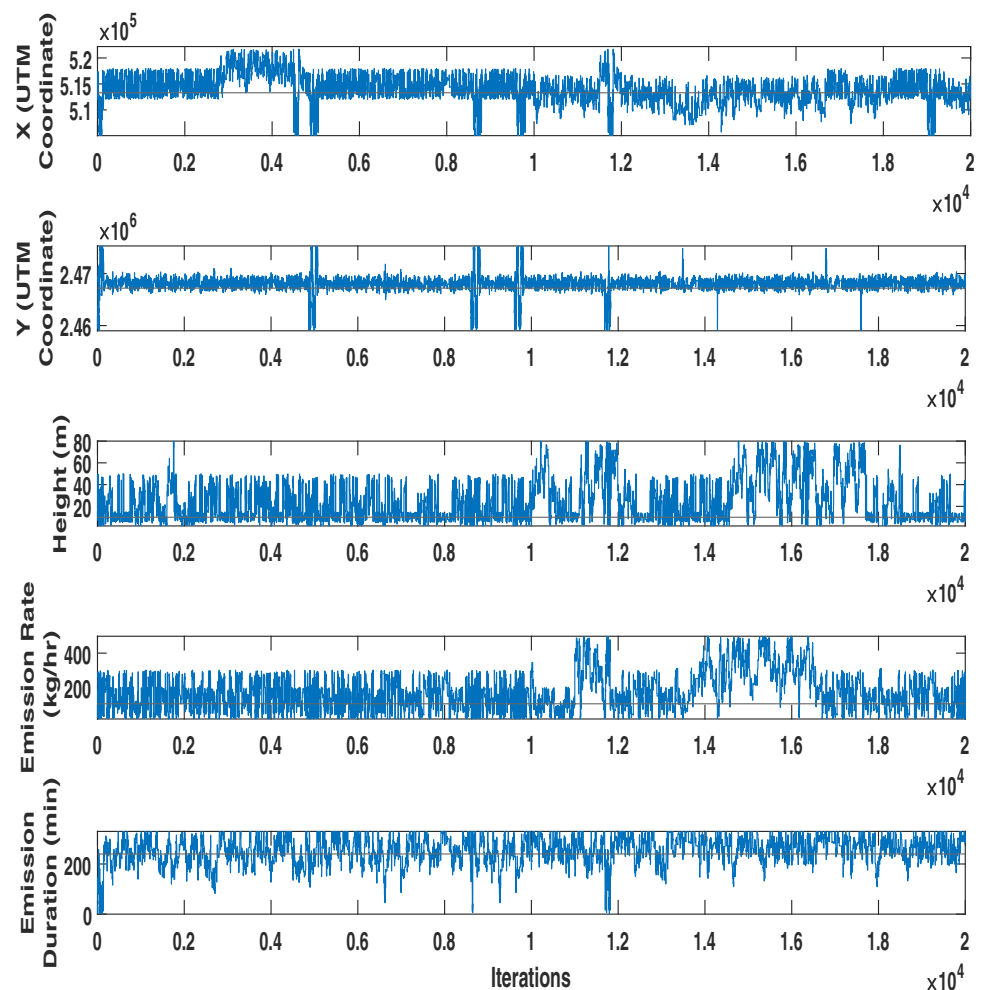
| i | Parameter, $M_i$        | Lower Bound | Upper Bound |
|---|-------------------------|-------------|-------------|
| 1 | x UTM Coordinate [m]    | 504,600     | 521,760     |
| 2 | y UTM Coordinate [m]    | 2,459,000   | 2,475,320   |
| 3 | Stack Height [m]        | 2           | 80          |
| 4 | Emission Rate [kg/h]    | 10          | 500         |
| 5 | Emission Duration [min] | 0           | 360         |

**Table 4** Settings of the numerical experiments conducted using the algorithm

| Experiment | Type of observation   | Time of observation [h] | Height of observation [m] | Used metric                  | Likelihood expression                     |
|------------|-----------------------|-------------------------|---------------------------|------------------------------|-------------------------------------------|
| 1          | $P$                   | 6                       | 10                        | $\chi$                       | Gaussian                                  |
| 2          | $F_{ic}$              | 6                       | 10                        | $\delta$                     | Exponential                               |
| 3          | $F$                   | 6                       | 10                        | $\delta$                     | Exponential                               |
| 4          | $F_{h_1}$ & $F_{h_2}$ | 6                       | $h_1 = 10; h_2 = 25$      | $\delta$                     | Exponential                               |
| 5          | $F_{T_1}$ & $F_{T_2}$ | $T_1=6; T_2=8$          | 10                        | $\delta$                     | Exponential                               |
| 6          | $F$ & $P$             | 6                       | $h_f = 25; h_p = 10$      | $\delta$ coupled with $\chi$ | Product of Exponential and Gaussian terms |

The information listed in columns 5, and 6 are inline with Section 3.2.  $P$  corresponds to a point-wise observation and  $F$  corresponds to a field observation. The subscript  $h$  represents the elevation,  $T$  represents an instant in time and  $ic$  refers to incomplete

**Fig. 6** Markov chains obtained in Exp. 3 for each of the five inferred parameters. The horizontal black lines indicate the reference values of each parameter



posterior distributions, the inverted solutions represented by the *MAP* are outlined in Table 5.

In Exp. 1, all inferred posterior distributions are flat with more than one high posterior probability region (indicated by the yellow zones in Fig. 7). This means that the distribution points out multiple sets of possible emission parameters, instead of a unique set of likely parameters. However, in Exp. 3, the inferred posterior probabilities of the parameters  $x$  and  $y$  (the source location) and  $d$  (the emission duration) exhibit relatively narrow distributions with their maxima at the reference values. The posterior probabilities of the other two parameters,  $h$  (the stack height) and  $q$  (the emission rate), show flatter distributions over their bounds. The maximum value for  $q$  is close to the reference value, unlike that of  $h$ . The contours of the high posterior probabilities are localized around restricted areas close to the reference values (Fig. 8).

The multiple high probability areas in Exp. 1 are misleading and difficult to interpret because there are no clear indications about the highest probability of a likely set of predicted emission parameters. Moreover, the *MAP* estimated the source as being 8.87 km from the true source in Exp. 1;

in comparison to only 2.9 km off in Exp. 3. These different solutions are related to the different global and local metrics used for quantifying the model-data dissimilarity. The local metric depends on a point-wise concentration value comparison; therefore, a deviation in the set of concentrations (spatially or temporally) will cause the double-penalty effect. The local discrepancy estimator double penalizes the inversion process, first by failing to predict the source parameters at their reference values and then by predicting these parameters away from their reference values. In this sense, the global Wasserstein metric is less discriminating [60]. Using such a metric in our study avoids the double-penalty effect and the multiple high posterior probability zones, providing more precise indications about the source parameters.

## 5.2 Treating discrete observations as an incomplete field

In Exp. 2, the point-wise observation is treated as an incomplete field observation to enable the use of the global Wasserstein metric. This successfully mitigated for the

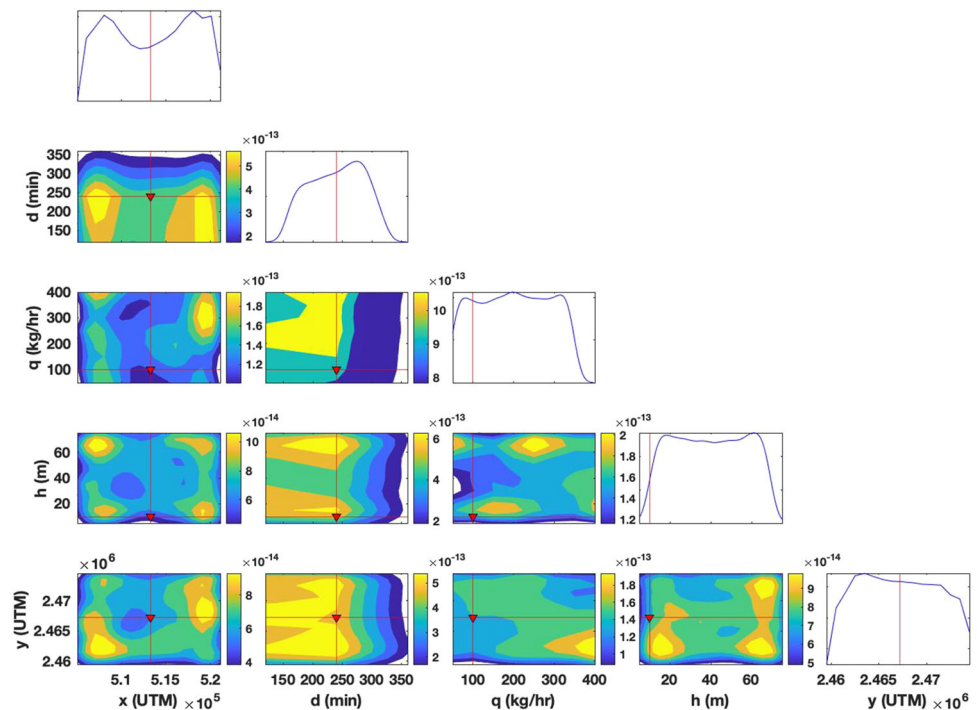
**Table 5** Inverted model parameters based on the maximum a posteriori estimator (*MAP*) where  $\zeta$  is the distance between the *MAP* and the true source

| Experiment       | $\zeta$ [km] | $h_{MAP}$ [m] | $q_{MAP}$ [kg/h] | $d_{MAP}$ [min] |
|------------------|--------------|---------------|------------------|-----------------|
| 1                | 8.87         | 15            | 300              | 240             |
| 2                | 3.27         | 20            | 150              | 120             |
| 3                | 2.9          | 8             | 100              | 240             |
| 4                | 1.53         | 10            | 100              | 240             |
| 5                | 0.5          | 12            | 100              | 240             |
| 6                | 2.22         | 10            | 100              | 240             |
| Reference Values | -            | 10            | 100              | 240             |

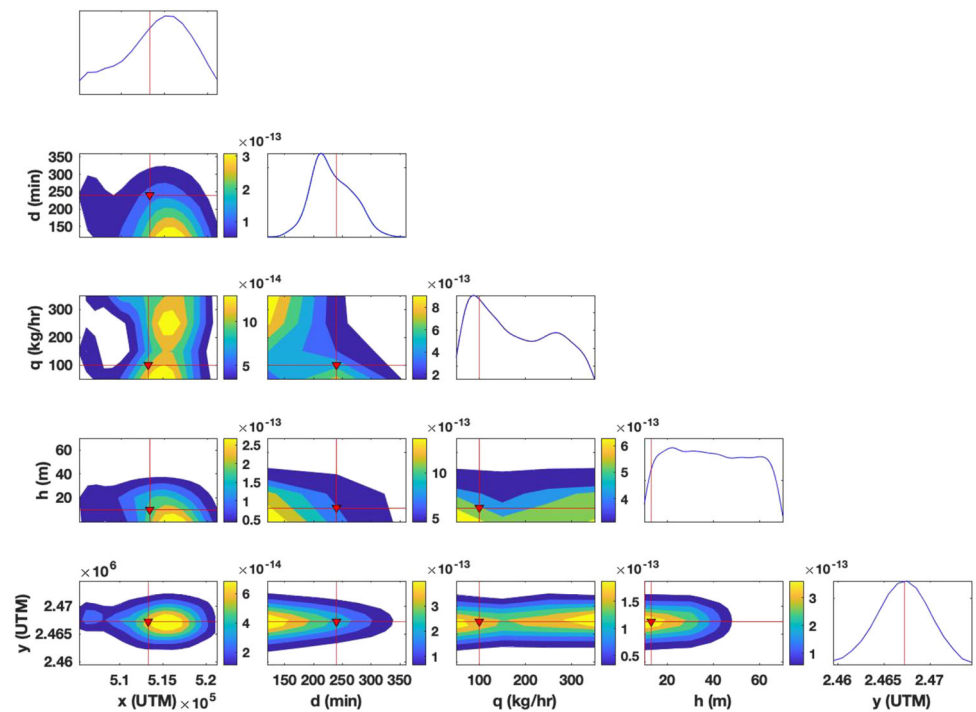
double-penalty effect observed in Exp. 1. In our analysis, we relate the double penalty effect with the appearance of multiple high posterior probability zones. Despite inverting flat distributions and wide contours, the high posterior probability zones were easier to interpret, as shown in Fig. 9. The *MAP* estimated the source location as being 3.27 km from the reference source location, which is comparable with the distance obtained in Exp. 3 using a complete field. In this context, the inverted solution yields a better prediction of the source location than that of Exp. 1. This may support considering the point-wise observation as an incomplete field in the context of inversion. However, the inverted solution predicts incorrect emission rate and emission duration (Table 5). This is because the metric is estimating the model-data misfit represented at only 20 points within the entire field; which is not sufficient to properly predict  $q$  and  $d$  as most of the concentration values are null.

### 5.3 Sensitivity to the observations

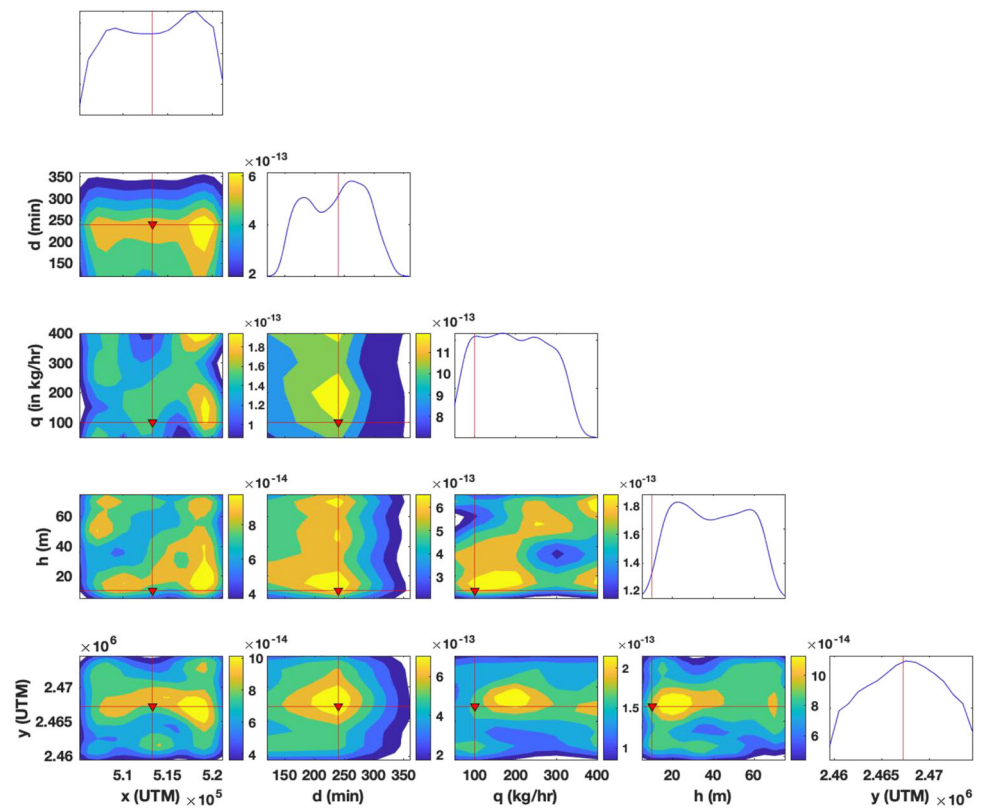
Exps. 4 and 5 used two concentration fields such that, in comparison with Exp. 3, one additional field observation is considered at a different elevation (in Exp. 4 at  $h = 25$  m) or at a different instant in time (in Exp. 5 at  $T = 8$  hours). The resulting Markov chains were generated with acceptance rates of 33% and 31%, respectively. To examine the benefit of additional observations, we also used kernel density estimates of the five-dimensional discrete posterior probability distribution by extracting 2D slices in the  $(x, y)$  parameter space, while fixing the values of the other three parameters  $h$ ,  $q$ , and  $d$  (Figs. 10, 11, 12 and 13). Note that only slices for  $d_{true} = 240$  min are represented because the posterior probabilities for emission durations away from that value are negligible. Also, we highlight that we are only observing 2D slices (i.e. a projections of the whole 5D posterior distribution), and

**Fig. 7** Corner plot of the obtained posterior probability distributions in Exp. 1. The red lines represent the reference values of each of the inferred parameters

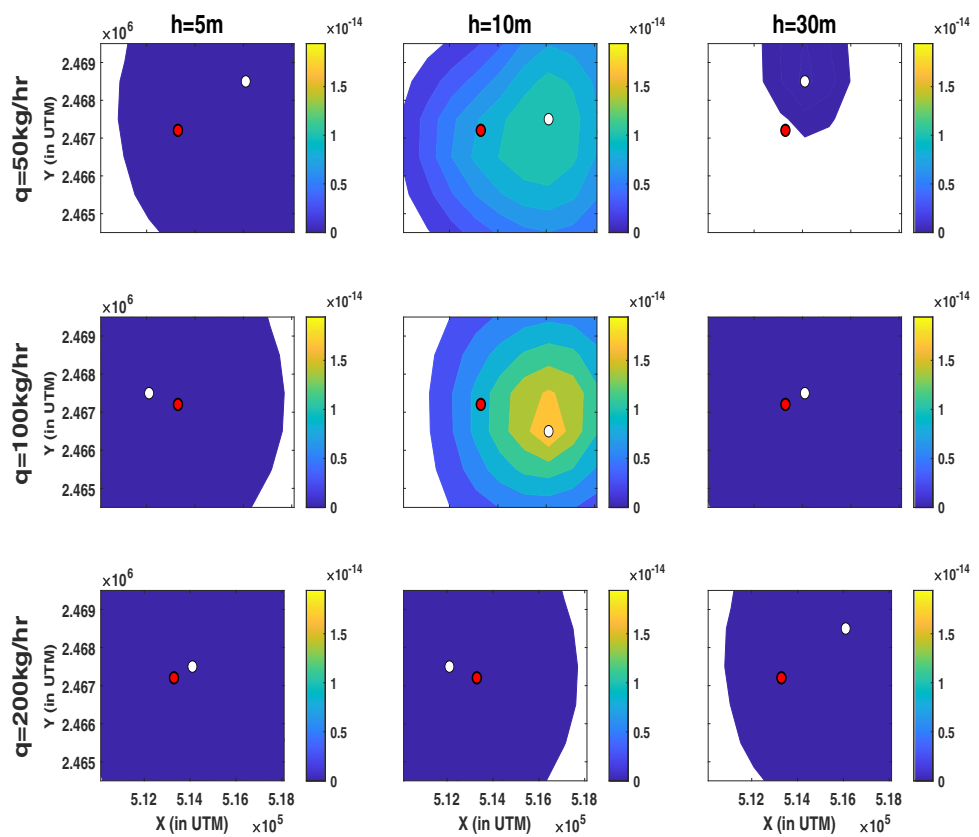
**Fig. 8** Corner plot of the obtained posterior probability distributions in Exp. 3. The red lines represent the reference values of each of the inferred parameters



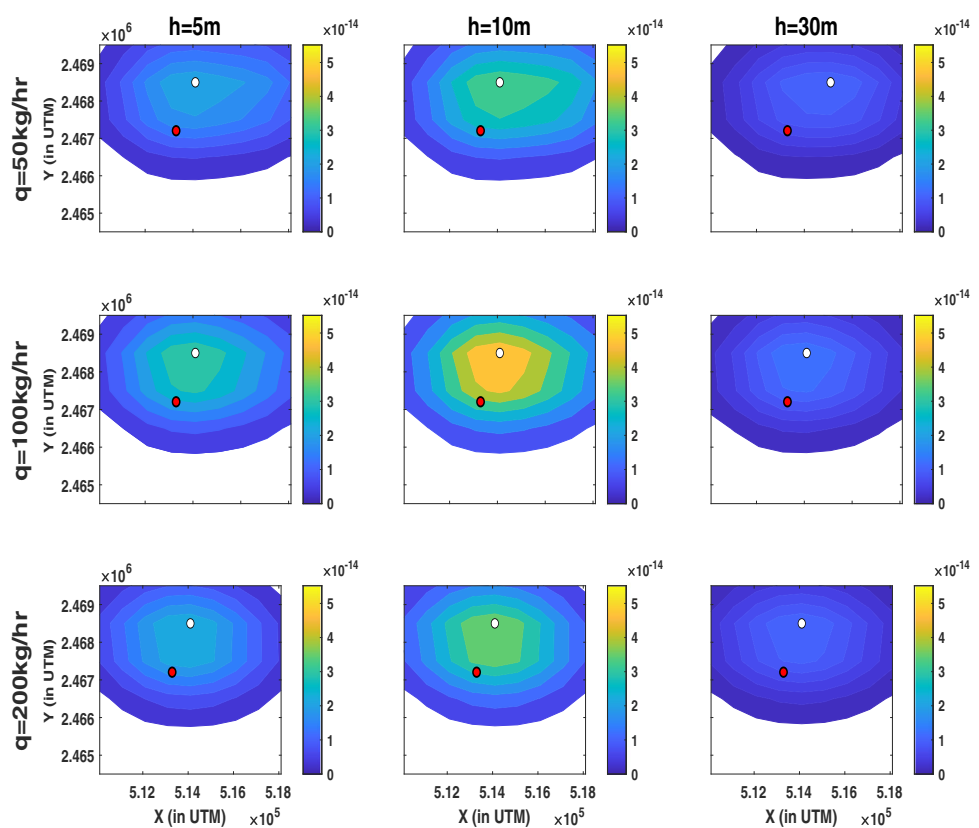
**Fig. 9** Corner plot of the obtained posterior probability distributions in Exp. 2. The red lines represent the reference values of each of the inferred parameters



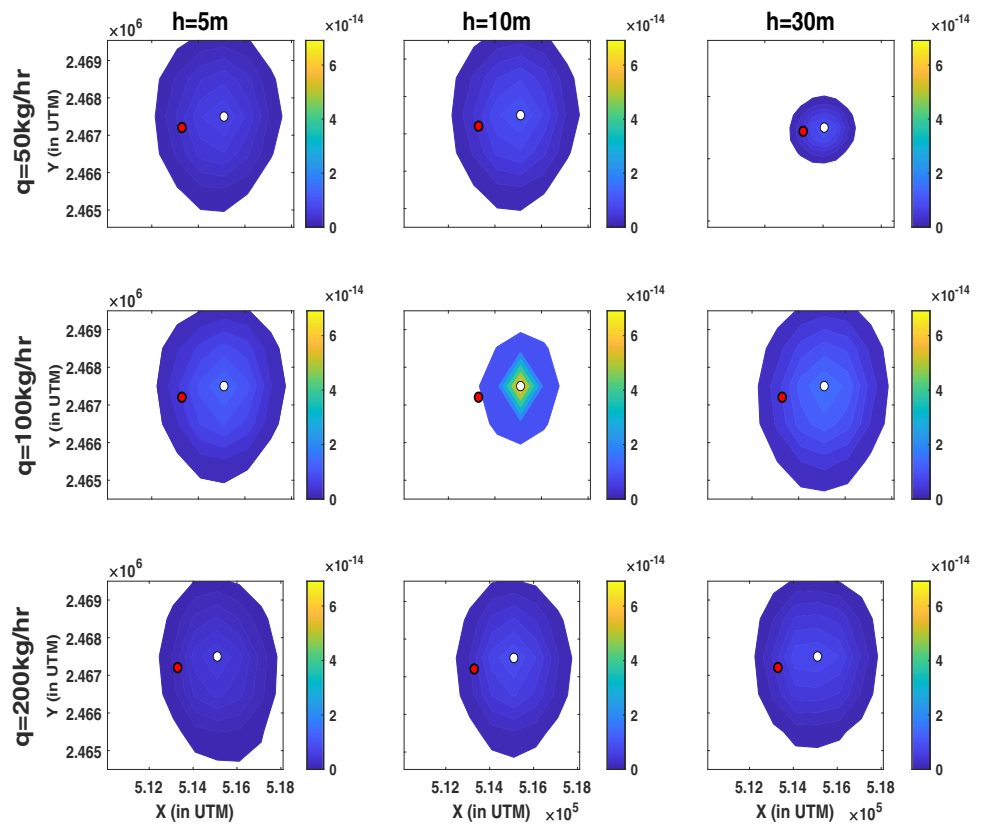
**Fig. 10** Two-dimensional (2D) slices of the posterior probability distribution in the  $(x, y)$  parameter space of Exp. 3 for  $d = 240$  min. The white dot in each slice represents the position at which the maximum posterior probability is obtained for each 2D map. The red dot represents the position of the true source. The reference values of  $h$  and  $q$  are 10m and 100kg/h, respectively



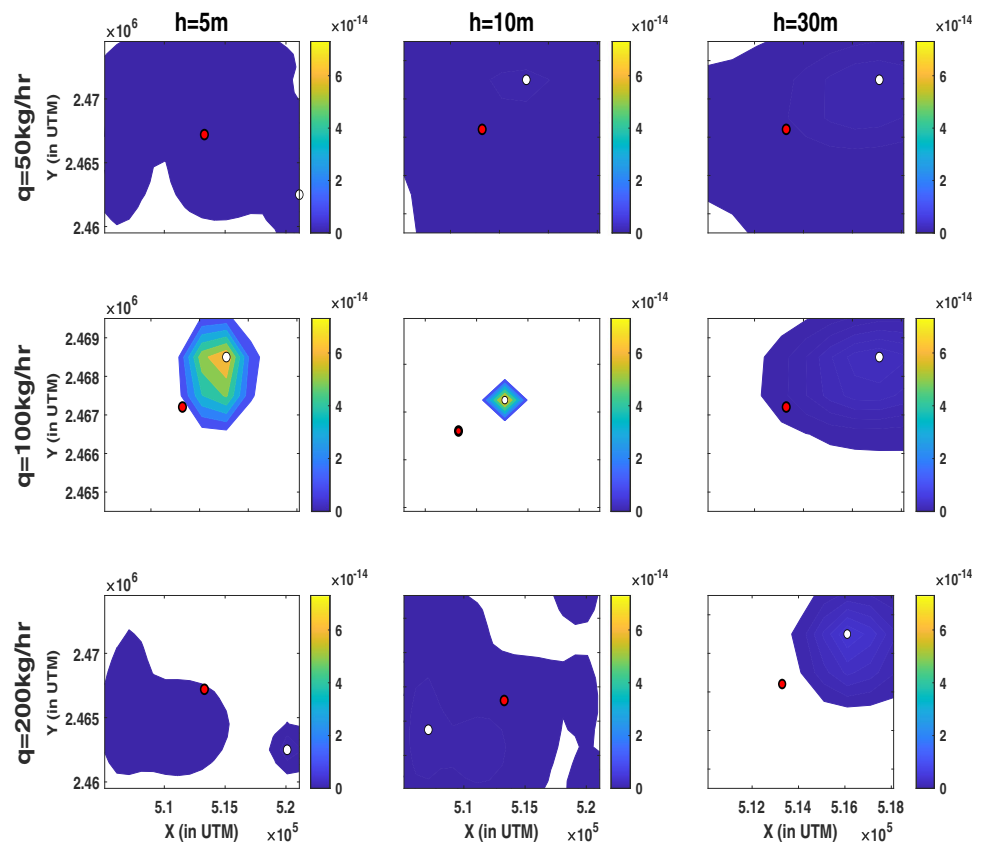
**Fig. 11** 2D slices of the posterior probability distribution in the  $(x, y)$  parameter space of Exp. 4 for  $d = 240$  min. The white dot in each slice represents the position at which the maximum posterior probability is obtained for each 2D map. The red dot represents the position of the true source. The reference values of  $h$  and  $q$  are 10m and 100kg/h, respectively



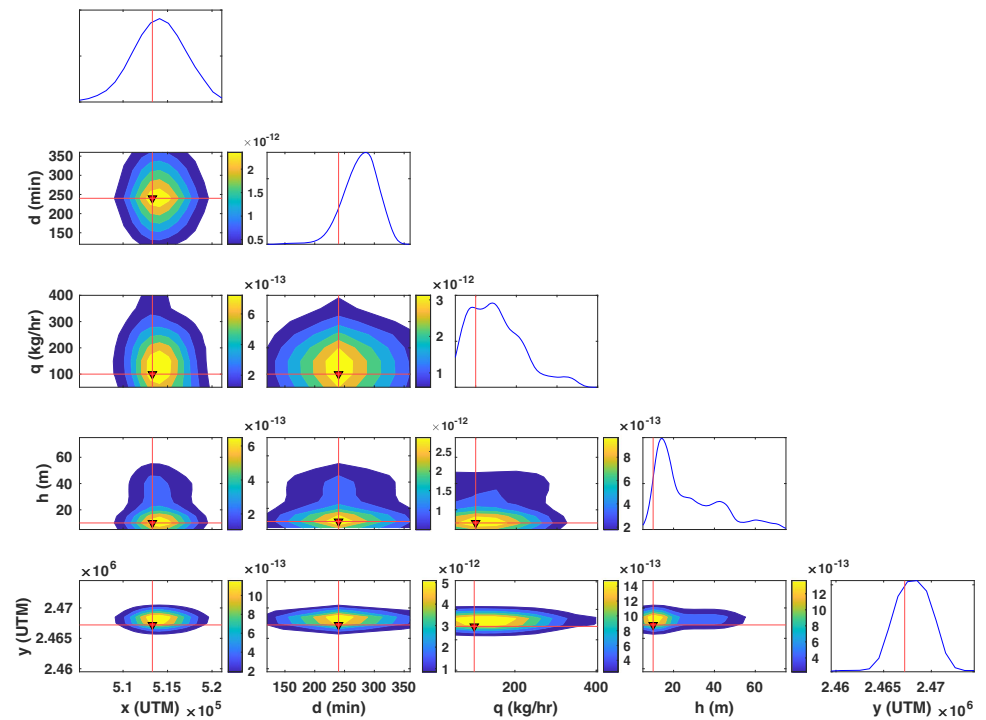
**Fig. 12** 2D slices of the posterior probability distribution in the  $(x, y)$  parameter space of Exp. 5 for  $d = 240$  min. The white dot in each slice represents the position at which the maximum posterior probability is obtained for each 2D map. The red dot represents the position of the true source. The reference values of  $h$  and  $q$  are 10m and 100kg/h, respectively



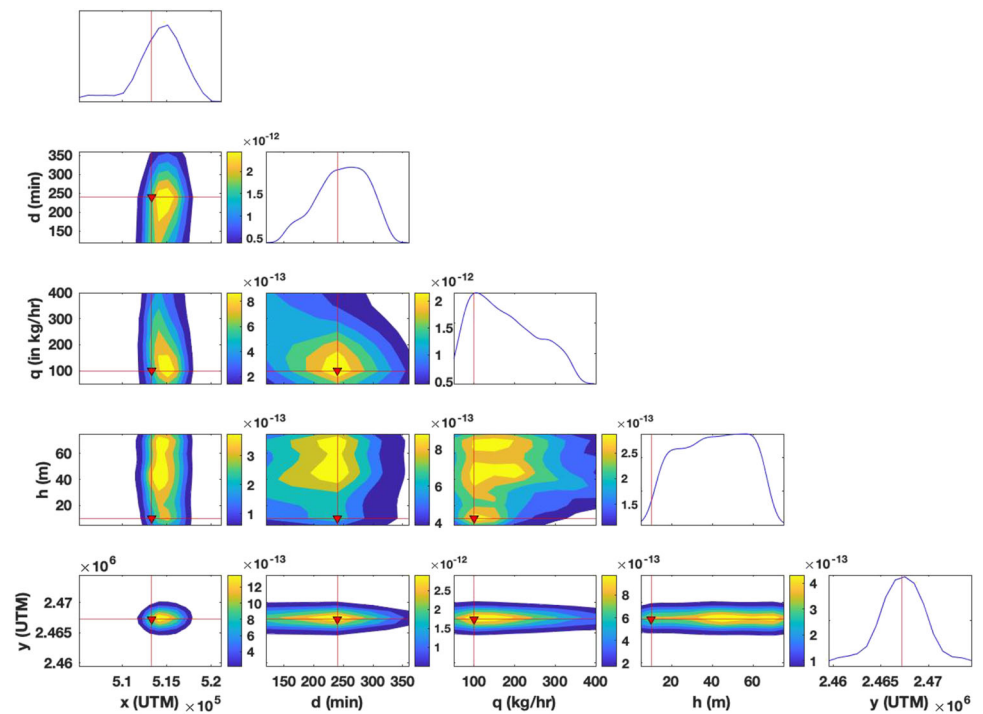
**Fig. 13** 2D slices of the posterior probability distribution in the  $(x, y)$  parameter space of Exp. 6 for  $d = 240$  min. The white dot in each slice represents the position at which the maximum posterior probability is obtained for each 2D map. The red dot represents the position of the true source. The reference values of  $h$  and  $q$  are 10m and 100kg/h, respectively



**Fig. 14** Corner plot of the obtained posterior probability distributions in Exp. 4. The red lines represent the reference values of each of the inferred parameters



**Fig. 15** Corner plot of the obtained posterior probability distributions in Exp. 5. The red lines represent the reference values of each of the inferred parameters



that the white dots in Figs. 10–13 originally have 5 different coordinates ( $x$ ,  $y$ ,  $h$ ,  $q$ ,  $d$ ), not just ( $x$ ,  $y$ ).

Through visual examination, one can notice that the highest posterior probability area in each experiment is located at the reference values of  $h$ ,  $q$ , and  $d$ . In addition, the sizes of the contours shrink when inferring for the two field observations of Exp. 5 as compared to Exp. 4 (Figs. 11 and 12); which means that the support of the posterior distribution is narrower. This suggests that more data, especially in time, reinforces the inversion process by localizing the bulk of the accepted samples of the Markov chains within a tighter area around the reference source. Moreover, the distance between the *MAP* and the true source decreased from 2.9 km in Exp. 3 to 1.53 km in Exp. 4 and 0.5 km in Exp. 5 (Table 5). Therefore, by using more informative observations, more confidence was gained through a posterior distribution that is indicative towards an inverted solution that is closer to the true values.

The effect of the additional field observations in Exps. 4 and 5 is further evaluated in the corner plots in Figs. 14 and 15. The improvement is clear in the shapes of the marginal distributions. In comparison to Exp. 3, the distributions exhibit a narrower support and their maxima are centered about the reference values. The inference of  $h$  further improves in Exp. 4 but not in Exp. 5. Therefore, a better inversion for  $h$  was achieved when using field observations available at different elevations not at at different times. This indicates that it is necessary to have information about the vertical profile of the concentrations in order to better invert for  $h$ . It also explains why it was hard to predict the true value of  $h$  in the previous experiments by just looking at horizontal profiles in the

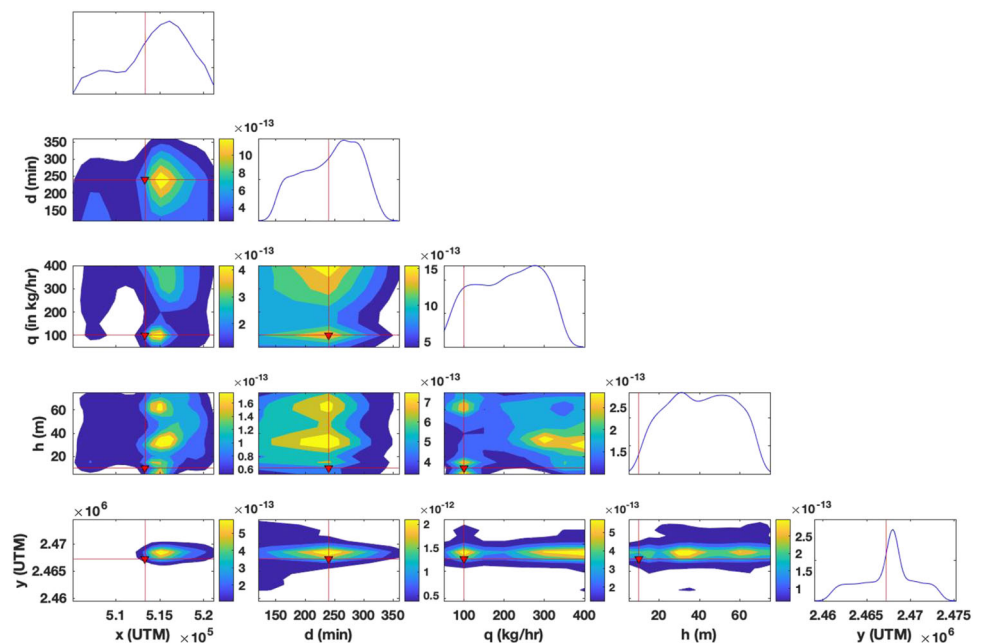
available observations. Furthermore, the joint distribution of  $q$  and  $d$  in Exp. 5 indicates that these two quantities are negatively correlated. This implies that the product of these two quantities can be predicted in the vicinity of a fixed value representing the emission mass.

Exp. 6 combines the two different types of observations, a field with point-wise concentration values. The subsequent inverted solution appears to be better than that obtained with only a field or a point-wise observation. The global metric reduced the double-penalty effect that would have been obtained by using only the local one. The corner plot in Fig. 16 indicates a single high posterior probability area in most of the subplots. We note that the spread contours are mostly blue, i.e. of negligible probabilities (as indicated by the color bar). So, the information conveyed in the point-wise observation made the distributions and the corresponding contours more informative of the emission parameters with a higher confidence. This effect is also clearly represented in Fig. 13 because the contours in the ( $x$ ,  $y$ ) plane at the reference values of  $h$ ,  $q$ , and  $d$  delineate a small area around the *MAP*, 2.22 km from the true source.

## 6 Summary and conclusions

Current advances in air quality monitoring provide a rich inventory of air pollution observations. Using these technologies to identify associated air contamination sources would allow local authorities to manage air pollution incidents and mitigate their harmful effects. Herein, we proposed a Bayesian inference framework that employs a Lagrangian

**Fig. 16** Corner plot of the obtained posterior probability distributions in Exp. 6. The red lines represent the reference values of each of the inferred parameters



dispersion model to infer the emission parameters in an urban environment. Numerical experiments were conducted to infer two types of observations: discrete concentration values and concentration fields. The inference algorithm involved a local discrepancy estimator in the experiments inferring discrete point observations, and a global optimal transport metric for the dissimilarity quantification in the experiments involving field observations. The results demonstrated the advantages of using a global metric as opposed to a local one by mitigating the double-penalty effect in the inverted solution. The global metric generated a posterior probability with lower uncertainties and hence a higher level of confidence in the inferred solution. Moreover, treating the discrete observations as an incomplete concentration field and using a global metric to evaluate its likelihood gave better prediction of the source location, than when the observations were considered as discrete data and their likelihood is evaluated by a local metric. Finally, the experiments revealed the sensitivity of the solution to the available observations, where less uncertainties in the inferred solution were associated with more informative observations; even when different types of observations were inverted. In this context, we found that observing more field concentrations in time provided the best solution.

High-resolution dispersion models coupled with Bayesian inference have been found to provide good inversion results of pollution sources in an urban environment. However, they require an additional computational cost that may not be feasible in a Bayesian framework. Therefore, we used an algorithm that defines the running domain of the model and hence limits computational costs. Alternatives that may reduce this high cost could be achieved using machine learning algorithms and polynomial chaos expansions. Another computationally demanding component of the proposed framework is estimating the global distance between two fields. Machine learning algorithms may also provide an efficient solution in that regard. These directions will be examined in our upcoming work.

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**Data Availability** The datasets generated during and analysed during the current study are available from the corresponding author on reasonable request.

## Declarations

**Conflicts of interest** The authors have no conflicts of interest to declare that are relevant to the content of this article.

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